

SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.

1.1 Problem Understanding

A multinational company is establishing a branch office in a rural location where high-speed wired Internet is not available. The branch must support cloud applications, video conferencing, communication with headquarters, and secure business operations.

The main constraints are:

- High-speed wired Internet is unavailable.
- The connectivity budget is limited.
- Cloud applications require reliable Internet access.
- Video conferencing requires sufficient bandwidth and low latency.
- Communication with headquarters must be secure.
- Business operations should continue even if the primary connection fails.
- The network should be scalable for future expansion.

1.2 Problem Decomposition

The problem can be divided into four major areas:

1. **Internet connectivity** – Select a suitable connection for the rural environment.
2. **Network infrastructure** – Select routers, switches, wireless access points, and other devices.
3. **Security** – Protect communication between the branch and headquarters.
4. **Reliability and scalability** – Provide backup connectivity and allow future expansion.

1.3 Proposed Solution

A 4G/5G fixed wireless connection can be used as the primary Internet connection, depending on cellular coverage. If reliable cellular coverage is unavailable, a satellite Internet connection can be used.

A suitable design is:

Internet → 4G/5G Router → Firewall/UTM → Switch → PCs, Servers and Access Points

A secondary Internet connection, such as another cellular provider or satellite connection, can be used as a backup.

1.4 Recommended Devices

Device	Purpose
4G/5G Router	Provides wireless Internet connectivity
Firewall/UTM	Controls traffic and protects the branch network
Layer 2/Layer 3 Switch	Connects computers, printers and other devices
Wireless Access Point	Provides Wi-Fi connectivity
VPN Gateway	Creates secure connection with headquarters
UPS	Keeps networking devices operational during short power failures
Network Monitoring Device/Software	Monitors connectivity and performance

1.5 Security Mechanisms

A **site-to-site VPN** should be established between the branch and headquarters. This encrypts business data travelling through the Internet.

Additional security mechanisms include:

- Firewall with access-control rules.
- WPA3 encryption for wireless networks.
- Strong administrator passwords.
- Multi-factor authentication for important services.
- Network segmentation using VLANs.

- Regular firmware and software updates.
- Intrusion detection/prevention where affordable.
- Secure DNS and web filtering.

1.6 Bandwidth Considerations

Cloud applications and video conferencing require considerably more bandwidth than basic email or web browsing. Therefore, the connection should be selected based on the expected number of employees and applications.

For example, if the branch has 30–50 employees, a high-quality 4G/5G connection can be suitable when sufficient coverage is available. Traffic prioritization can be implemented using **Quality of Service (QoS)**.

QoS can give higher priority to:

1. Video conferencing.
2. Voice communication.
3. Cloud business applications.
4. Normal web browsing.
5. Software updates and background downloads.

This prevents non-critical traffic from consuming all available bandwidth.

1.7 Reliability

Because uninterrupted business operation is required, a backup connection is important.

Primary: 4G/5G fixed wireless

Backup: Second cellular provider or satellite connection

The router can automatically switch to the backup connection when the primary link fails. A UPS can also keep the router, firewall and switch running during short power interruptions.

1.8 Cost and Scalability

The proposed solution avoids expensive wired infrastructure because it uses existing wireless infrastructure. Initially, the company can deploy a moderate-capacity router, firewall and switch.

For future expansion:

- Additional switches can be added.
- More access points can be installed.
- VLANs can be created for new departments.
- Bandwidth can be upgraded.

- Additional VPN connections can be configured.
- A stronger firewall can be deployed when the branch grows.

1.9 Justification

The solution balances the major constraints effectively.

Constraint	Solution
No wired high-speed Internet	4G/5G or satellite
Limited budget	Wireless connection avoids expensive cabling
Video conferencing	Adequate bandwidth + QoS
Secure communication	Site-to-site VPN
Reliability	Backup Internet connection + UPS
Security	Firewall, VLANs, WPA3 and MFA
Future growth	Scalable switches and IP architecture

1.10 Conclusion

A 4G/5G-based network with a backup connection, firewall, VPN, VLANs and QoS is a practical solution for the rural branch. It provides reasonable bandwidth without requiring expensive wired infrastructure. The backup connection improves availability, while VPN and firewall technologies protect communication with headquarters. The architecture can also be expanded as the company grows.

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

2.1 Problem Understanding

The university needs campus-wide wireless access for more than 3,000 students across classrooms, laboratories, libraries and hostels.

The main constraints are:

- More than 3,000 users.
- Large number of simultaneous connections.
- Limited number of wireless access points.
- Large physical coverage area.
- Signal interference must be minimized.
- Students and staff require secure authentication.
- Network performance must remain acceptable.
- The available budget is limited.

The main challenge is therefore to achieve **high user capacity with fewer access points** without creating excessive interference.

2.2 Problem Decomposition

The problem can be divided into:

1. **Coverage planning** – Determine suitable AP locations.
2. **Capacity planning** – Support many simultaneous users.
3. **Frequency planning** – Reduce wireless interference.
4. **Authentication** – Secure access.
5. **Traffic management** – Prioritize important traffic.
6. **Centralized management** – Monitor and configure APs efficiently.

2.3 Proposed Wireless Architecture

A centralized wireless architecture can be used:

Internet → Core Router/Firewall → Core Switch → PoE Switches → Wireless Access Points

A **Wireless LAN Controller (WLC)** or centralized cloud-managed WLAN system can manage the access points.

The university can divide the wireless network into different SSIDs/VLANs:

- **University-Staff**
- **University-Student**
- **University-Guest**

- **University-IoT**

This prevents different categories of users from directly accessing sensitive resources.

2.4 Access Point Placement

APs should not simply be installed at equal distances. They should be placed according to **user density and building structure**.

Classrooms

Access points should be placed in corridors or strategically within classroom blocks. Areas containing many students should receive higher AP density.

Laboratories

Laboratories require higher capacity because many computers and mobile devices may be connected simultaneously.

Library

The library requires strong capacity because students may use laptops, tablets and smartphones simultaneously.

Hostels

Hostels should have APs distributed across floors and residential blocks rather than relying on a single powerful AP.

A basic placement strategy is:

High-density areas → More AP capacity

Low-density areas → Fewer APs

This makes better use of the limited AP budget.

2.5 Frequency Band Selection

Modern dual-band or tri-band access points should be used.

5 GHz

The 5 GHz band should be preferred for high-performance users because it provides more usable channels and generally experiences less interference than 2.4 GHz.

It is suitable for:

- Video conferencing.
- Online classes.

- Cloud applications.
- High-speed downloads.
- Laboratory activities.

2.4 GHz

The 2.4 GHz band can be retained for devices that have limited 5 GHz support and for devices requiring greater range.

However, excessive use of 2.4 GHz should be avoided because it has fewer non-overlapping channels and is more susceptible to interference.

6 GHz

If the selected equipment and regulations support Wi-Fi 6E/7 operation, the 6 GHz band can be used in appropriate areas for newer devices and additional capacity.

2.6 Reducing Interference

Interference can be reduced by:

- Performing a wireless site survey.
- Selecting appropriate channels.
- Reducing unnecessary AP transmission power.
- Avoiding excessive AP overlap.
- Using automatic channel management.
- Separating heavily used APs appropriately.
- Using modern Wi-Fi standards.
- Monitoring channel utilization.

The objective should not be maximum signal strength everywhere. Instead, the goal is **adequate coverage with sufficient capacity and controlled interference.**

2.7 Authentication and Security

For students and staff, **WPA2-Enterprise or preferably WPA3-Enterprise** can be used with a RADIUS authentication server.

The authentication process can be:

Student Device → Wi-Fi AP → RADIUS Server → University Authentication Database

This provides individual user authentication rather than a single shared password.

For guests, a separate guest network can be created with Internet-only access.

Security mechanisms include:

- WPA3-Enterprise.
- 802.1X authentication.
- RADIUS server.
- VLAN segmentation.
- Guest network isolation.
- Firewall policies.
- Strong identity management.
- Regular monitoring.

2.8 Capacity Management

Because the university has more than 3,000 students, simply increasing signal range is not enough. Each AP has a practical limit on the number of active clients.

Therefore:

- High-density areas should receive high-capacity APs.
- Client load should be distributed between APs.
- Band steering should encourage compatible devices toward higher-capacity bands.
- QoS should prioritize educational applications.
- Bandwidth limits can be applied to guest users.

2.9 Cost Optimization

Since the university has a limited budget, the design should avoid placing expensive APs everywhere.

A better strategy is:

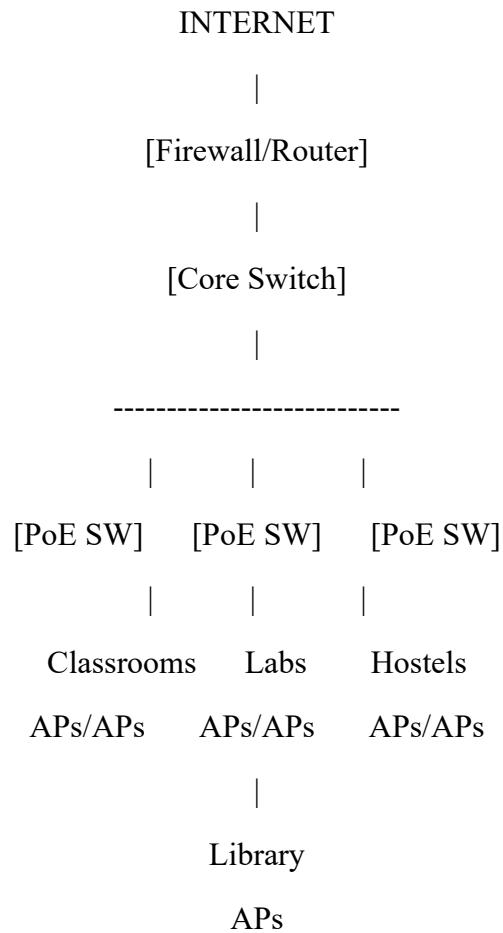
High-density zones → High-capacity APs

Medium-density zones → Standard APs

Low-density zones → Minimal AP deployment

Centralized management also reduces operational costs because administrators can configure and monitor APs from one location.

2.10 Proposed Network Structure



2.11 Constraint-Based Justification

Constraint	Design Decision
3,000+ students	High-capacity Wi-Fi APs
Limited APs	Capacity-based AP placement
Interference	Channel planning and power control
Security	WPA3-Enterprise + 802.1X
Different users	Student/Staff/Guest VLANs
High performance	5 GHz/6 GHz preferred
Limited budget	APs concentrated in high-density areas
Future growth	Wi-Fi 6E/7-ready infrastructure

2.12 Conclusion

The university should deploy a **centrally managed, capacity-based wireless network** instead of simply maximizing coverage. APs should be concentrated in high-density locations such as classrooms, laboratories and libraries. 5 GHz and, where appropriate, 6 GHz should be preferred for high-performance users. WPA3-Enterprise, 802.1X and RADIUS provide secure authentication. This design balances **coverage, capacity, security, interference control and cost**.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

3.1 Problem Understanding

A financial institution handles highly sensitive customer and banking information. The network must therefore provide strong security while maintaining good performance and controlling operational costs.

The major constraints are:

- Sensitive financial information must be protected.
- Employees need access to authorized customer information.
- Online banking services must remain available.
- Strict security and compliance requirements must be followed.
- Network performance should not be significantly reduced.
- Operational costs should remain manageable.
- Unauthorized access and attacks must be detected quickly.

3.2 Problem Decomposition

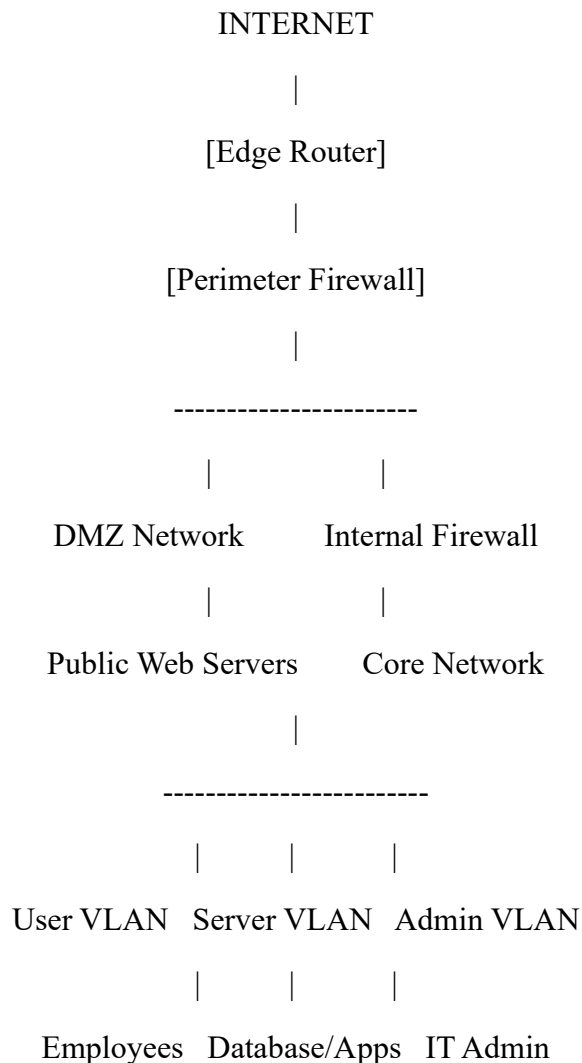
The problem can be divided into:

1. **Network segmentation**
2. **Firewall deployment**
3. **Authentication and authorization**

4. **Intrusion detection and prevention**
5. **Secure Internet access**
6. **Monitoring and logging**
7. **Performance and cost optimization**

3.3 Proposed Secure Architecture

A **layered defense architecture** should be implemented.



The architecture separates public-facing systems from internal financial systems.

3.4 DMZ

Public services such as web servers should be placed in a **Demilitarized Zone (DMZ)**.

For example:

- Public website.
- Internet banking front-end servers.

- Public DNS services.
- External application gateways.

The DMZ prevents Internet-facing servers from having unrestricted access to the internal network.

If an Internet-facing server is compromised, segmentation limits the attacker's ability to directly reach internal databases.

3.5 Network Segmentation

VLANs should separate different types of systems.

Example:

VLAN	Purpose
VLAN 10	Employee systems
VLAN 20	Finance applications
VLAN 30	Database servers
VLAN 40	IT administration
VLAN 50	Guest devices
VLAN 60	Security/Monitoring

Access between VLANs should be controlled through firewall rules or Layer 3 security policies.

For example:

Guest VLAN → Internet: Allowed

Guest VLAN → Internal Database: Denied

Employee VLAN → Authorized Application Servers: Allowed

Employee VLAN → Database Directly: Denied

This implements the principle of **least privilege**.

3.6 Firewall Deployment

A next-generation firewall should be placed at the network perimeter.

The firewall can provide:

- Stateful packet inspection.
- Application control.
- Access-control rules.

- VPN services.
- Intrusion prevention.
- Web filtering.
- Malware detection.

For highly sensitive environments, an additional internal firewall can separate critical financial systems from ordinary employee networks.

3.8 Intrusion Detection and Prevention

An **IDS/IPS** should monitor network traffic for suspicious activity.

IDS

An Intrusion Detection System identifies suspicious behavior and generates alerts.

IPS

An Intrusion Prevention System can actively block or prevent known malicious traffic.

The system should monitor:

- Repeated login failures.
- Port scanning.
- Suspicious network connections.
- Malware traffic.
- Unauthorized access attempts.
- Abnormal data transfers.

Security logs should be collected centrally so that administrators can investigate incidents.

3.9 Data Protection

Sensitive customer information should be protected both during transmission and storage.

Methods include:

- TLS for web applications.
- VPN for remote connections.
- Encryption of sensitive databases.
- Secure backups.
- Access logging.
- Data-loss prevention where required.

Employees should only receive access to information necessary for their job role.

3.10 Performance Considerations

Strong security should not unnecessarily slow down business operations.

Performance can be maintained by:

- Using appropriately sized firewalls.
- Applying security policies efficiently.
- Separating critical traffic using VLANs.
- Using load balancing for public applications.
- Monitoring bandwidth utilization.
- Avoiding unnecessary inspection of low-risk internal traffic where policy permits.
- Regularly reviewing firewall rules.

Security devices should be selected according to expected throughput and concurrent connection requirements.

3.11 Cost Optimization

A financial institution does not necessarily need separate physical security appliances for every security function.

A suitable **next-generation firewall/UTM** can combine several functions, such as:

- Firewall.
- VPN.
- IDS/IPS.
- Application control.
- Web filtering.

Virtualization can also be used for selected internal services where appropriate.

Centralized monitoring reduces administrative effort and helps identify problems before they become major incidents.

3.12 Constraint-Based Justification

Constraint	Design Decision
Sensitive customer information	Encryption + segmentation
Strict security requirements	Firewall + IDS/IPS + MFA

Employee access	Role-based access control
Public online banking services	DMZ
Unauthorized access	802.1X + MFA
Performance	Proper firewall sizing + traffic management
Compliance	Centralized logging and access control
Limited operational budget	Integrated security platforms
Future threats	Scalable IDS/IPS and monitoring

3.14 Conclusion

The financial institution should implement a **defense-in-depth architecture** using DMZs, VLAN segmentation, firewalls, MFA, 802.1X, role-based access control and IDS/IPS. Public-facing services should be isolated from internal systems, while sensitive databases should be placed in a highly restricted network segment. The solution provides strong security without unnecessarily affecting network performance and can be implemented using integrated security platforms to control costs.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured	Breaks problem into sub-parts with	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
		sub-problems effectively	some logical flow		
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand